The limiting nonunimodular-root ratio for $P_{m,1}$

A solution to MathDB 353931

Principia Math In Progress

18 September 2026

Abstract

We prove Stankov's conjecture that the limit ratios associated with $P_{m,1}$ tend to approximately 0.209. Starting from the published fixed-m root-counting formula, a cellwise comparison proves convergence with error at most $2/(3m)$. The limit is $2\big(\mathrm{Li}_2(1/2) - \mathrm{Li}_2(-1/2)\big)/\pi^2 = 0.20885431502703392737\ldots$.

Task 1: the original problem and its full solution

Write $\Phi_m(x) = \sum_{j=0}^{m-1} x^j$. The family in the source is

$$P_{m,1}(x,y) = x^{m-1}\Phi_m(x) + x^{m-1}y + \Phi_m(x)y^2.$$

For integers $m \geq 2$ and $n \geq 1$, set

$$\begin{aligned} H_{n,m}(x) &= \Phi_m(x)\big(x^{2n+m-1}+1\big) + x^{n+m-1} \\ &= x^{n+m-1}\left(1 + \sum_{j=0}^{m-1}(x^{n+j} + x^{-n-j})\right). \end{aligned}$$

This is the source's $H_{2n+2m-2}$, of degree $2n+2m-2$, and $P_{m,1}(x, x^{n+m-1}) = x^{m-1}H_{n,m}(x)$. Let $C(H)$ be the number of roots of H off the unit circle, counted with multiplicity, divided by $\deg H$, and put

$$L_m = \lim_{n\to\infty} C(H_{n,m}).$$

The fixed-m limit and its following measure representation are *prior results of Stankov*, not claims of this note:

$$L_m = \frac{1}{\pi}\,\mathrm{meas}\{t \in [0,\pi] : 2|\sin(mt)| < \sin t\}. \tag{1}$$

See Theorem 1 and the proof of Theorem 9 in the published paper [1], especially page 429. Its envelope is $\sin(mT/2)/\sin(T/2)$ and its central coefficient is 1; substitution $T = 2t$ gives (1). The monomial factor x^{m-1} changes the root count and degree by a fixed amount for fixed m, so it does not change this limit. Thus L_m is precisely the $LC(P_{m,1})$ of Conjecture 11 on page 431 (Conjecture 2.1 in the arXiv version). The conjecture asks whether L_m converges to approximately 0.209.

Theorem 1. *For every integer $m \geq 2$,*

$$|L_m - L| \leq \frac{2}{3m}, \tag{2}$$

where

$$L = \frac{4}{\pi^2}\int_0^{\pi/2} \arcsin\left(\frac{\sin t}{2}\right)\,dt \tag{3}$$

$$= \frac{2}{\pi^2}\big(\mathrm{Li}_2(1/2) - \mathrm{Li}_2(-1/2)\big) = \frac{2}{\pi^2}\sum_{k=0}^{\infty}\frac{4^{-k}}{(2k+1)^2}. \tag{4}$$

In particular, the conjectured limit exists and equals

$$L = 0.208854315027033927371594356326472240306 5\ldots.$$

Proof. Define $h(s) = 2\arcsin(s/2)/\pi$ for $0 \le s \le 1$ and $g(t) = h(\sin t)$ for $0 \le t \le \pi$. Partition $[0,\pi]$ into the m intervals

$$I_j = [j\pi/m, (j+1)\pi/m], \qquad 0 \le j < m.$$

For a constant threshold $s \in [0,1]$, the substitution $u = mt - j\pi$ shows that

$$\frac{m}{\pi}\,\mathrm{meas}\{t \in I_j : 2|\sin(mt)| < s\} = h(s). \tag{5}$$

Indeed, the set of $u \in [0,\pi]$ with $2\sin u < s$ has length $2\arcsin(s/2)$, also when $s = 0$.

Let u_j and v_j be respectively the minimum and maximum of $\sin t$ on I_j. Monotonicity in the threshold and (5) give

$$h(u_j) \le A_j := \frac{m}{\pi}\,\mathrm{meas}\{t \in I_j : 2|\sin(mt)| < \sin t\} \le h(v_j).$$

The cell average $B_j = (m/\pi)\int_{I_j} g(t)\,dt$ lies in the same interval. Consequently

$$\left|L_m - \frac{1}{\pi}\int_0^{\pi} g(t)\,dt\right| \le \frac{1}{m}\sum_{j=0}^{m-1}|A_j - B_j|$$

$$\le \frac{1}{m}\sum_{j=0}^{m-1}\big(\max_{I_j} g - \min_{I_j} g\big) \le \frac{\mathrm{Var}_{[0,\pi]}\, g}{m}.$$

Here each oscillation is at most the variation on its cell, and variations add over a partition. The function g increases from 0 to $h(1) = 1/3$ on $[0,\pi/2]$ and decreases to 0 on $[\pi/2,\pi]$. Its total variation is therefore $2/3$. This proves (2); reflection about $\pi/2$ gives (3). It also proves convergence without assuming independence of t and mt or interchanging the two root-counting limits.

It remains to evaluate the integral. For $0 \le a \le 1/2$, let $J(a) = \int_0^{\pi/2}\arcsin(a\sin t)\,dt$. The derivative of the integrand is uniformly bounded by $2/\sqrt{3}$ on this rectangle, so differentiation under the integral is justified. For $0 < a \le 1/2$, substitution $u = \cos t$ gives

$$J'(a) = \int_0^1 \frac{du}{\sqrt{1 - a^2 + a^2u^2}} = \frac{\operatorname{arctanh} a}{a} = \sum_{k=0}^{\infty}\frac{a^{2k}}{2k+1}.$$

The last series converges uniformly for $0 \le a \le 1/2$. Since $J(0) = 0$, termwise integration yields

$$J(a) = \sum_{k=0}^{\infty}\frac{a^{2k+1}}{(2k+1)^2} = \frac{\mathrm{Li}_2(a) - \mathrm{Li}_2(-a)}{2}, \qquad \mathrm{Li}_2(z) = \sum_{r=1}^{\infty}\frac{z^r}{r^2}.$$

Putting $a = 1/2$ proves (4). □

Endpoints and numerical value. All sets above are measured by Lebesgue length, so shared cell endpoints have no effect. Equality in the threshold also has measure zero: the analytic functions $2\sin(mt) \pm \sin t$ have only finitely many zeros on this compact interval and are not identically zero. The positive series in (4) has the explicit tail bound

$$0 < \sum_{k=N}^{\infty} \frac{4^{-k}}{(2k+1)^2} \leq \frac{4^{1-N}}{3(2N+1)^2}.$$

The accompanying script uses $N = 80$ and rational bounds on π from Machin's formula to certify the displayed forty decimal digits. Numerical root-location checks are supplementary only.

Task 2: the sharp rate of convergence

Determine the optimal order of $|L_m - L|$, and the leading term of $L_m - L$. In particular, establish or refute an $O(m^{-2})$ bound and decide whether even and odd m have different leading coefficients. The estimate in Theorem 1 does not answer these questions.

Task 3: monotonicity and finite-m inequalities

Determine whether L_m is strictly increasing for every $m \geq 2$ and whether $L_m < L$ always holds. Seek explicit finite-m bounds that improve both the source's estimates and (2).

Task 4: a variable central coefficient

Replace x^{n+m-1} in $H_{n,m}$ by λx^{n+m-1}, with $\lambda \geq 0$. Determine the outer limit as a function of λ, including the transition when the threshold $\lambda \sin t/2$ exceeds 1. Obtain exact expressions and regularity statements at the transition.

Task 5: simultaneous growth of the two parameters

Find quantitative conditions on $n = n(m)$ ensuring that $C(H_{n(m),m}) \to L$. This requires bounds for the inner root-counting limit that are uniform in m. Investigate the regime where n/m tends to a finite positive constant as well as the regime $n/m \to \infty$.

Verification and attribution. The fixed-m formula (1) is credited to [1]; related general counting and averaging methods occur in [2, 3, 4]. This note proves the particular outer limit, its explicit value, and (2). The documented bounded search found no prior full claim for that target. All four followups are proposed questions, not claimed results. Review was by the authoring agent only.

References

[1] D. Stankov, *The number of nonunimodular roots of a reciprocal polynomial*, Comptes Rendus Mathématique **361** (2023), 423–435. doi:10.5802/crmath.422. Earlier version: arXiv:2203.07710v1 (2022).

[2] D. W. Boyd and M. J. Mossinghoff, *Small limit points of Mahler's measure*, Experimental Mathematics **14**(4) (2005), 403–414, Section 6. doi:10.1080/10586458.2005.10128936.

[3] D. Stankov, *The alternative to Mahler measure of a multivariate polynomial*, arXiv:2502.02803v1 (2025).

[4] D. Stankov, *Approximation of the number of nonunimodular zeros of a self-reciprocal polynomial*, arXiv:2509.01015v1 (2025).